

## Human-Computer Interaction

### *Who is the user? – assessment of sex, age and emotional state based on the appearance.*

Human faces convey a wealth of information – not only about a person's identity, but also about their gender, approximate age, mood, and even their level of engagement or attention. In the context of Human-Computer Interaction (HCI), the ability of a computer system to detect and interpret human facial features enables the creation of more natural and adaptive interfaces. Systems that can recognize whether a user is male or female, estimate their age, or detect emotions such as happiness, surprise, anger, or sadness can be used in a variety of applications (in service kiosks, information systems, or educational environments).

Today, numerous freely available tools and applications allow users to experiment with facial detection and emotion recognition technologies without the need for advanced programming skills or expensive equipment. These tools make it possible to conduct simple experiments and observe how algorithms interpret human appearance, revealing both their potential and their limitations.

Below are examples of available solutions that can be used for this laboratory exercise (but the selection is not limited to them – students are encouraged to explore and choose other freely available tools or demos):

- Mobile applications

Face Shape Detector (Android)

<https://play.google.com/store/apps/details?id=com.detection.faceshape>

– detects faces and analyzes basic facial features directly on a smartphone.

- PC software

SHORE™ – Fraunhofer Institute for Integrated Circuits IIS

<http://www.iis.fraunhofer.de/en/bf/bsy/download/shore.html>

– downloadable demo software for face detection and emotion recognition, developed by one of the leading research institutes in computer vision.

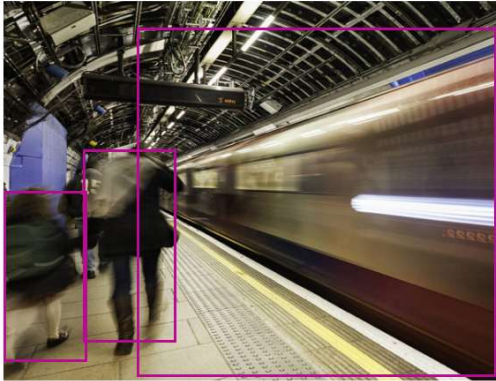
- Web-based tools and online demos

Microsoft Azure Face API Demo

<https://azure.microsoft.com/en-us/products/ai-services/ai-vision>

– allows testing of cloud-based algorithms for facial analysis.

Zobacz, jak to działa



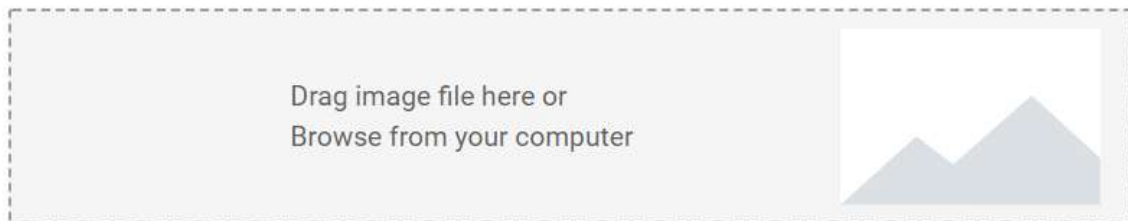
| NAZWA FUNKCJI: | WARTOŚĆ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| Obiekty        | [ { "rectangle": { "x": 93, "y": 178, "w": 115, "h": 237 }, "object": "person", "confidence": 0.764 }, { "rectangle": { "x": 0, "y": 229, "w": 101, "h": 206 }, "object": "person", "confidence": 0.624 }, { "rectangle": { "x": 161, "y": 31, "w": 439, "h": 423 }, "object": "subway train", "parent": { "object": "train", "parent": { "object": "Land vehicle", "parent": { "object": "Vehicle", "confidence": 0.926 }, "confidence": 0.923 }, "confidence": 0.917, "confidence": 0.801 } ] |
| Tagi           | [ { "name": "train", "confidence": 0.9975446 }, { "name": "platform", "confidence": 0.9955431 }, { "name": "station", "confidence": 0.979800761 }, { "name": "indoor", "confidence": 0.9277198 }, { "name": "subway", "confidence": 0.8389395 }, { "name": "clothing", "confidence": 0.5043765 }, { "name": "pulling", "confidence": 0.4317162 } ]                                                                                                                                              |

## Google Cloud Vision API

<https://cloud.google.com/vision>

– a cloud-based computer vision service by Google that provides pre-trained models for face detection, emotion recognition, image labeling, and other vision-related tasks, accessible directly through a web interface or API.

Try the API



And many others:

<https://mymoodscan.ai/>

<https://labs.everypixel.com/demo>

<https://imentiv.ai/>

<https://skybiometry.com/>

<https://www.faceplusplus.com/>

**Note:** Remember to use only free versions or demo modes of the listed tools. Do not provide any payment details or credit card information. If a particular tool requires payment or registration beyond a free tier, simply choose another available solution. The goal of this exercise is to test freely accessible technologies.

## Exercise 1 - Evaluation of face detection software

- Find 3 solutions of software for face detection (web services, demo versions, open source solutions, etc.).
- Familiarize with the principles of operation of each solution.

- Conduct experiments for several sample face images and indicate the advantages and disadvantages of each solution.
- Are the results obtained by different solutions same?
- Are all faces detected correctly?
- Check operation in complicated lighting conditions. Examples:



- Present results in graphical form and add your own comments.

### **Exercise 2 - Evaluation of software for age and sex of the user assessment**

- Find 3 solutions of software for age and sex recognition (web services, demo versions, open source solutions, etc.).
- Conduct experiments for several examples of face images in each of the selected solutions. Present the results (preferably in tabular form). Do not forget to present test photos.
- Are the results obtained by different solutions convergent? Try to explain.
- Would you decide on any of the tested solutions to determine automatically the age of the user of your application?

### **Exercise 3 - Evaluation of software for human emotions assessment based on facial expressions**

- Find 3 solutions of software for human emotion assessment (web services, demo versions, open source solutions, etc.).
- Familiarize with the principles of operation of each solution.
- Check what kind of emotions are possible to detect using each solution. Compare functionality.
- Conduct experiments for several sample face images in each of the selected solutions.
- Present, compare and evaluate the results obtained.
- What features in the appearance of the face are responsible for assigning faces to the appropriate category of emotions?

**Report**

- Prepare a report on the exercises performed in PDF format.
- Send the report to: [anowosielski@zut.edu.pl](mailto:anowosielski@zut.edu.pl)
- In the e-mail title, please use the scheme:
  - HCI Your Full Name